

STEPGEN · REQUIREMENTS

StepGen requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/stepgen		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

StepGen requirements

These are first-pass requirements for the concept. Targets are proposals for review and will be checked by calculation at TRL 3 and revised after co-design with users.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Accept the mechanical power a person can sustain	50 to 100 W nominal at the pedals (70 kg user, 15 cm step, 30 to 60 steps per minute); drivetrain and generator rated for 250 W peak	Work-per-step calculation; component ratings
R2	Store a useful amount of energy per hour of stepping	35 Wh or more at 30 steps per minute and 60 Wh or more at 60 steps per minute, for the nominal user	Efficiency chain calculation; later metered bench test
R3	Comfortable step height	Adjustable 10 to 20 cm at the foot, nominal 15 cm; pedal top no higher than 30 cm above the floor	Linkage geometry in the model
R4	Resistance suits a wide range of users	5 or more selectable resistance levels; users of 40 to 120 kg can step at 30 to 60 steps per minute without bottoming out; display shows level, steps, W and Wh	Load-line calculation per user mass; design review
R5	Charge the PowerBox correctly	Output 39.0 to 54.6 V DC (13S pack at 46.8 V nominal, assumed; to confirm against PowerBox), charge current limited to 3 A, charging stops when the pack is full or the PowerBox signals a fault	Charger specification; interface review with PowerBox
R6	Efficient energy chain	65 % or more from pedal work to energy stored in the pack at nominal load	Loss budget calculation; later measurement
R7	Safe extra-low voltage	Every conductor under 60 V DC in all conditions, including generator overspeed to 3,000 rpm and a disconnected load	Generator constant (Kv) and overspeed calculation
R8	Quiet indoor use	60 dBA or less at 1 m while stepping at 60 steps per minute	Component noise estimate; later sound level measurement
R9	Fits a small room and can be moved	Footprint 0.9 x 0.7 m or less, height 1.3 m or less, mass 35 kg or less, movable by two people	Massing model; mass estimate
R10	Guard all moving parts	Chains, sprockets, belt, flywheel and generator fully enclosed; no drivetrain part reachable by hand or foot; crush gaps at the pedals of at least 50 mm for toes and 25 mm for fingers, or fully closed	Design review against guarding practice (ISO 13854 and ISO 13857)
R11	Safe for untrained users	Handrail on both sides at about 1.0 m above the floor; non-slip pedal treads; elastomer end stops so a pedal cannot drop hard if the controller fails; frame stable with a 120 kg user leaning on one rail; emergency stop that brakes the flywheel	Stability calculation; design review
R12	Low cost and buildable	Parts cost \$400 or less; built with common hand tools, a drill press and basic welding or bolted steel; bicycle parts where possible	Priced BOM

■ Assumptions

- Mechanical work per step equals body weight times step height: $70 \text{ kg} \times 9.81 \text{ m/s}^2 \times 0.15 \text{ m} = \text{about } 103 \text{ J per step}$.
- A step is one pedal going down; 30 to 60 steps per minute is 0.5 to 1 step per second.
- The PowerBox pack is a 13-series lithium-ion pack (46.8 V nominal, about 39 to 54.6 V). The PowerBox design must confirm this.
- Users step at a cadence they choose; the load controller sets the resistance so the pedal descends at a comfortable speed for that user's weight.

■ Requirements at risk

- **R2 at the low end and the pitch figure.** The nominal user at 30 steps per minute stores about 36 Wh per hour, only just above the 35 Wh target. The figure of 60 to 100 Wh per hour in the current pitch is reached only at the top of the cadence band or by heavier users; see the precis and review note.
- **R8** is unverified. Chain and belt noise at the generator speed is not yet estimated.
- **R12** is met with almost no margin (about \$397 of \$400).